Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Solving Systems Using Elimination and Substitution **Date:** __________

Advanced (Habanero)

Q1. Solve the system using substitution: $x + y = 4$, $x - y = 2$
(A) $x = 3, y = 1$
(B) $x = 2, y = 2$
(C) $x = 1, y = 3$
(D) $x = 4, y = 0$
(E) $x = 0, y = 4$

Q2. Solve the system using elimination: $2x+3y = 7$, $x-2y = -3$
(A) $x = 1, y = 2$
(B) $x = 2, y = 1$
(C) $x = 3, y = 0$
(D) $x = 0, y = 3$
(E) $x = -1, y = -2$

Q3. What is the solution to the system: $x + y = 4$, $x + y = 5$
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Cannot be determined
(E) None of the above

Q4. Solve the system using substitution: $2x-3y = 1$, $x+2y = 3$
(A) $x = 2, y = 1$
(B) $x = 1, y = 2$
(C) $x = 3, y = 0$
(D) $x = 0, y = 3$
(E) $x = -1, y = -2$

Q5. What is the solution to the system: $x-2y = 3$, $3x-6y = 9$
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Cannot be determined
(E) None of the above

Q6. Solve the system using elimination: $x+2y = 5$, $3x-2y = 7$
(A) $x = 3, y = 1$
(B) $x = 2, y = 1$
(C) $x = 1, y = 3$
(D) $x = 4, y = 0$
(E) $x = 0, y = 4$

Q7. What is the solution to the system: $2x + 2y = 4$, $x + y = 2$
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Cannot be determined
(E) None of the above

Q8. Solve the system using substitution: $3x-y = 2$, $x+4y = 11$
(A) $x = 2, y = 1$
(B) $x = 1, y = 2$
(C) $x = 3, y = 0$
(D) $x = 0, y = 3$
(E) $x = -1, y = -2$

Open Ended Questions (FRQ)

Q9. Solve the system using elimination: $2x + 5y = 17$, $x - 3y = -7$

Q10. Solve the system using substitution: $x - 2y = -1$, $3x + 4y = 10$

Chef's Tips

- Tip 1: Use substitution or elimination to solve the systems
- Tip 2: Check for one solution, no solution, or infinitely many solutions
- Tip 3: Use algebraic methods to solve the systems